CHARACTERISATION OF THE NORMALIZATION BY A UNIVERSAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. Read in full under the Lead's 12:4x rule (screening hit re-read, pb 2239B). The author writes verbatim 'We omit the verification that the diagram commutes.' - a routine commutativity check for the constructed morphism h : X' -> Z; the construction of h and the subsequent argument are given in full.

Allowed standard prerequisites (editorial).

- Equivalence between quasi-coherent sheaves of algebras and affine morphisms. Lemma \ref{lemma-affine-equivalence-algebras}, cited at proof lines 13969-13975. Provides the morphism h and its uniqueness.
- Integral and birational morphisms are described affinely by integral extensions of coordinate rings. Lemma \ref{lemma-normalization} and Lemma \ref{lemma-birational-integral}, used at proof lines 13957-13968 and 13988-13999. Reduce the universal property to the affine case.

Source reference. Original lemma 035I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4129–4129. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 035H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4128–4128. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2482–2482. Navigation pointer to a statement the

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 035F: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4126–4126. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01SA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2331–2331. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 03GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4529–4529. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01KV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2064–2064. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 035D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4124–4124. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. The factorization $f = \nu \circ f'$, where $\nu : X' \to X$ is the normalization of X in Y is characterized by the following two properties:*

(1) *the morphism ν is integral, and*
(2) *for any factorization $f = \pi \circ g$, with $\pi : Z \to X$ integral, there exists a commutative diagram*

$$\begin{array}{ccc} Y & \xrightarrow{g} & Z \\ {\scriptstyle f'}\downarrow & \nearrow_{h} & \downarrow{\scriptstyle \pi} \\ X' & \xrightarrow{\nu} & X \end{array}$$

for some unique morphism $h : X' \to Z$.

Moreover, the morphism $f' : Y \to X'$ is dominant and in (2) the morphism $h : X' \to Z$ is the normalization of Z in Y.

Proof. Let $\mathcal{O}' \subset f_*\mathcal{O}_Y$ be the integral closure of $\mathcal{O}_X$ as in Definition 035H. The morphism ν is integral by construction, which proves (1). Assume given a factorization $f = \pi \circ g$ with $\pi : Z \to X$ integral as in (2). By Definition 01WH π is affine, and hence Z is the relative spectrum of a quasi-coherent sheaf of $\mathcal{O}_X$-algebras $\mathcal{B}$. The morphism $g : Y \to Z$ corresponds to a map of $\mathcal{O}_X$-algebras $\chi : \mathcal{B} \to f_*\mathcal{O}_Y$. Since $\mathcal{B}(U)$ is integral over $\mathcal{O}_X(U)$ for every affine open $U \subset X$ (by Definition 01WH) we see from Lemma 035F that $\chi(\mathcal{B}) \subset \mathcal{O}'$. By the functoriality of the relative spectrum Lemma 01SA this provides us with a unique morphism $h : X' \to Z$. We omit the verification that the diagram commutes.

It is clear that (1) and (2) characterize the factorization $f = \nu \circ f'$ since it characterizes it as an initial object in a category.

From the universal property in (2) we see that f' does not factor through a proper closed subscheme of X'. Hence the scheme theoretic image of f' is X'. Since f' is

quasi-compact (by Schemes, Lemma 03GI and the fact that ν is separated as an affine morphism) we see that $f'(Y)$ is dense in X'. Hence f' is dominant.

Observe that g is quasi-compact and quasi-separated by Schemes, Lemmas 01KV and 03GI. Thus the last statement of the lemma makes sense. The morphism h in (2) is integral by Lemma 035D. Given a factorization $g = \pi' \circ g'$ with $\pi' : Z' \to Z$ integral, we get a factorization $f = (\pi \circ \pi') \circ g'$ and we get a morphism $h' : X' \to Z'$. Uniqueness implies that $\pi' \circ h' = h$. Hence the characterization (1), (2) applies to the morphism $h : X' \to Z$ which gives the final assertion of the lemma. □